Continual Reinforcement Learning for Microalgae Bioreactor Control: Outperforming Static and PID Systems in Carbon Capture

Sia Gupta

ABSTRACT

Microalgae are microorganisms that fix CO_2 through photosynthesis at high rates. With atmospheric CO_2 levels exceeding 420 ppm, carbon capture technologies are increasingly needed. Unlike terrestrial plants, microalgae can grow in non-arable land and be converted into biofuels. However, most photobioreactors rely on static control strategies that are unable to adapt to changing growth dynamics. This project implemented reinforcement learning to regulate light and nutrients in a photobioreactor to improve biomass productivity compared to static and PID control systems. A Proximal Policy Optimization (PPO) reinforcement learning agent was trained in a simulation using the Monod kinetics for nutrient-limited growth and a Haldane inhibition model for light response. The agent measured dissolved oxygen and elapsed time, continuously adjusting LED intensity and nutrient injection. The reward function optimized dissolved oxygen as a proxy for photosynthetic performance. A continual learning approach allowed the agent to tune its policy in real time on a photobioreactor growing Chlorella vulgaris, and its performance was compared with static and PID control systems under identical inoculum conditions across independent 5-day cultivation cycles. The experimental bioreactor achieved a mean 10.2% higher dissolved oxygen accumulation and 94.7% greater final biomass compared to the static baseline. Compared to PID control, the RL system achieved 4.3% higher dissolved oxygen and 42.8% greater biomass. This study shows that continual reinforcement learning can significantly improve microalgae growth, suggesting a potential improvement to biofuel production and carbon capture strategies. Further experiments are recommended to optimize temperature and scale up to full-size photobioreactors for real-world validation.

INTRODUCTION

This study aims to adress the growing concerns of climate change by implementing a continual reinforcement learning system to regulate both light intensity and nutrient injection in a Chlorella vulgaris photobioreactor.

Climate change has been a long-standing issue that has accelerated in recent years, with atmospheric CO2 levels increasing from ~280ppm pre industrialization to over 420ppm (Mulhern, 2020). It poses an urgent threat to global ecosystems, destroying biodiversity and long-term ecological stability. Traditional methods to combat climate change include carbon sequestration strategies, such as afforestation or soil carbon storage, yet require significant land areas and are often limited by geographic, environmental, or socio-economic constraints (Islam, 2025). These limitations emphasize the need for alternative carbon capture approaches that are both space-efficient and environmentally adaptable.

Microalgae cultivation in photobioreactors has emerged as a promising alternative. Microalgae can fix CO2 through photosynthesis at rates exceeding those of terrestrial plants, with some systems removing as much CO2 as an acre of trees while occupying only a fraction of the space (Arif Gasilov, 2025). In addition, algae biomass can be repurposed into biofuels, bioplastics, and agricultural feedstocks, holding economic as well as environmental value. In fact, microalgae produce oil at ~11 times the yield per hectare of the next best alternative, palm oil (Chisti, 2007). Moreover, algae bioreactors, systems that cultivate algae for biomass production, can operate in non-arable areas and have the potential to contribute to net-zero carbon emissions.

Despite these advantages, current algae bioreactors face several challenges. Most rely on static, human-defined controls for variables such as CO2 input, light intensity, temperature, and nutrients. These fixed parameters are often unable to respond to dynamic environmental conditions or complex system interactions. Prior research has explored a few adaptive approaches. PID controllers can respond to sensor feedback, but they follow fixed mathematical rules and cannot improve from experience (Luca et al., 2018).

Fuzzy logic controllers are more flexible, but they remain rule based with no learning component (Luca et al., 2018). While some recent advances employ artificial intelligence and machine learning to improve bioreactor performance, these approaches are generally predictive or supervised, requiring large pre-collected datasets and lacking the ability to continuously adapt in real time (Zocco, 2025). Addressing these limitations is crucial for realizing and utilizing the full potential of algae-based carbon capture.

This study addresses that gap by implementing a continual reinforcement learning system to regulate both light intensity and nutrient injection in a Chlorella vulgaris photobioreactor. Continual reinforcement learning is an approach in which the agent never stops updating its policy from live data, unlike traditional models that freeze after training. A Proximal Policy Optimization (PPO) agent was first trained in a simulation using Monod kinetics and Haldane light inhibition, then deployed on embedded hardware where it continued learning from real sensor feedback every 24 hours. Its performance was compared against both static control and a PID controller across independent five-day trials. Chlorella vulgaris was selected as the model species due to its rapid growth rate and high carbon capture efficiency (Garodia et al., 2025).

In the study the central research question was: how can an embedded reinforcement learning algorithm autonomously regulate environmental conditions in an algae bioreactor to maximize carbon capture efficiency by continuously learning? There were three primary engineering goals: (1) design a simulation to converge to a positive reward within 200,000 training steps, (2) deploy a continual learning mechanism to autonomously update the policy on real hardware, and (3) achieve a statistically significant improvement over both static and PID with a minimum 10% increase in mean dissolved oxygen and 20% increase in biomass over the static.

METHODOLOGY

Materials

This study created reactor units from 16 oz mason jars that were each equipped with a Vernier Go Direct Optical Dissolved Oxygen probe, a WS2812B LED strip, and a micro-servo-controlled syringe pump for nutrient injections. All reactors were controlled by a Raspberry Pi 4B using Visual Studio Code. The nutrient medium was a F/2 concentrate diluted in distilled water growing Chlorella vulgaris (from Algae Research Supply). The enclosure was constructed from cardboard. Additional materials included silicon tubing, 20 ml syringes, wires, and breadboards. Main software dependencies were Python 3.11, Stable Baselines 3, Gymnasium, PyTorch, NumPy, Pandas, and Matplotlib.

Prototype Design & Iterations

Simulation Iterations

Throughout the development, there were three main iterations across two tracks: simulation environment and RL algorithm.

The first iteration used the Droop model to simulate growth. The Droop model tracks the internal cell quota, or the intracellular nutrient concentration per cell.

Equation 1: Droop Model

$$\mu = \mu_{max}\left(1 - \frac{Q_{min}}{Q}\right)$$

This results in high biological accuracy, yet it requires measurement of parameters that were inaccessible at lab scale. Without these parameters, the model could not be tuned, making it impractical.

The RL algorithm in this stage used Augmented Random Search (ARS). ARS works by randomly adjusting policy parameters, testing whether those adjustments improve the reward, and updating the policy in the direction that worked better. This approach works well for simple policies with single output; however, the bioreactor action space requires two continuous outputs: light intensity and nutrient injection. These are coupled through the reward function because changing light without also adjusting the nutrients produces a different outcome than changing both together. ARS could not find a consistent update direction. Random adjustments prevented the policy from stabilizing on a strategy that balanced both variables effectively.

The second iteration replaced Droop with Andrews model, which handled substrate injection by modeling how growth rate peaks at optimal substrate concentration and then declines at higher concentrations:

Equation 2: Andrews Model

$$\mu = \frac{\mu_{max} S}{K_s + S + \frac{S^2}{K_i}}$$

Here, μ is the specific growth rate, μmax is the maximum growth rate, S is the substrate concentration, Ks is the half-saturation constant, and Ki is the inhibition constant. The S^2/K_i accounts for the growth rate to decrease at excessively high substrate concentrations. However, Andrews only models nutrient substrate and does not include any light term, leaving the agent without any awareness.

ARS was continued for this version.

The third and final iteration combined Monod kinetics for nutrient-limited growth with Haldane light inhibition and replaced ARS with Proximal Policy Optimization (PPO).

Monod kinetics model how nutrient concentration drives growth rate. As nutrients increase, growth increases up to a maximum then plateaus:

Equation 3: Monod

$$\mu = \mu_{max} \cdot \frac{S}{K_{sN} + S}$$

Here, μ is the specific growth rate, μmax is the maximum growth rate (0.15d-1), S is the nutrient concentration, K_sN is the half-saturation constant (3.0). Unlike Andrews, Monod does not model inhibition at high nutrient concentrations, but this was fine as nutrient toxicity concerns were handled by setting a maximum daily safety cap.

Haldane light inhibition extends Monod by adding a light-dependent term that suppresses photosynthesis at high intensities:

Equation 4: Haldane

$$P = P_{max} \cdot \frac{I}{I + K_{sL} + \frac{I^2}{K_i}}$$

where P is the photosynthetic rate, Pmax is the maximum rate (150.0), I is light intensity, KsL is the light half-saturation constant (20.0), and Ki is the inhibition constant (80.0). When combined with Monod kinetics, the full oxygen production rate used in the simulation became:

Equation 5: Combined O2 Rate

$$O_2 = P_{max} \cdot \frac{I}{I + K_{sL} + \frac{I^2}{K_i}} \cdot \frac{S}{K_{sN} + S}$$

The reward function used to train the agent was:

Equation 6: Reward Function

$$R = \frac{O_2 rate}{5.0} - (I \times 0.005) + \begin{cases} 10 & \text{if} O_2 > 60 \\ 0 & \text{otherwise} \end{cases}$$

This penalized excessive light usage while also rewarding high oxygen production.

ARS was replaced with Proximal Policy Optimization. PPO corrects the instability of ARS by updating the policy using gradient descent instead of random adjustments. Meanwhile it also limits how large each update can be through a clipped objective function:

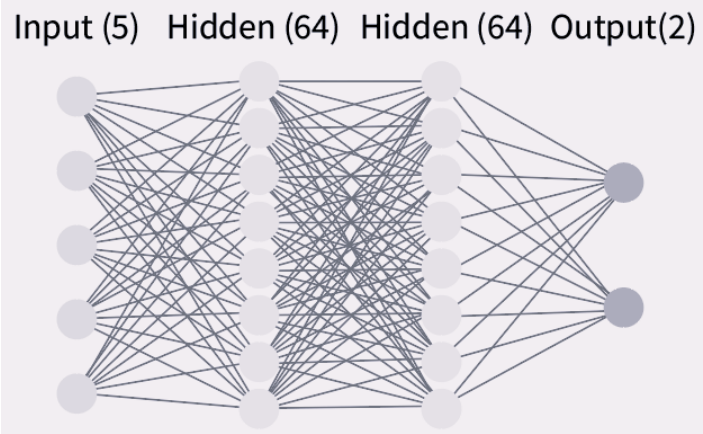
Equation 7: PPO clipped objective

$$L = \mathbb{E}[\min\left(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) A_t\right)]$$

where rt is the ratio between the new and old policy probabilities, At is the advantage estimate representing how much better and action was than expected, and ϵ is the clipping threshold (0.2). The clipping mechanism prevents the policy from making large updates, making PPO suitable for biological reward systems. The agent was trained for 200,000 steps using a multilayer perceptron (MLP) policy network with two hidden layers of 64 neurons each and ReLU activations.

A continual learning mechanism was added to run a gradient update using live sensor data every 48 decision steps. In this, it updated its neural network weights based on actual outcomes rather than simulated predictions.

Figure 1: Neural network architecture diagram



Hardware Iterations

The first hardware iteration was a small reactor with no visible separation between reactors.

It was also made out of thin balsa wood and was extremely delicate. Here light from the RL reactor strip bleed into the environment and additional environmental factors were contaminating the light variable. Since light intensity is the primary control action, any cross contamination would invalidate the comparison.

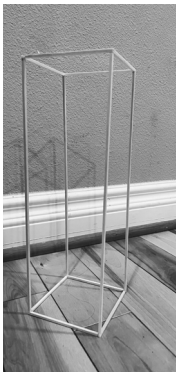
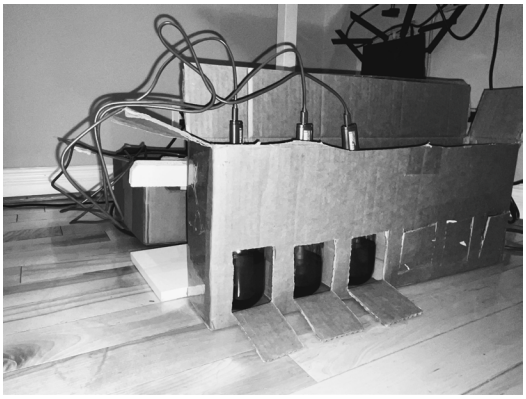


Figure 2: Photo of first hardware iteration

The second iteration fixed this by constructing a cardboard enclosure divided into isolated compartments, one per reactor.

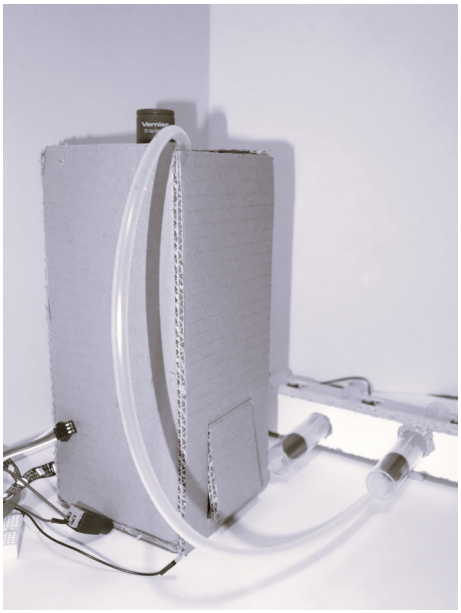
Figure 3: Photo of cardboard enclosure showing isolated compartments.



This prevented light bleeding between conditions. Each compartment was connected to a race platform containing dual stools for the syringe tubing running from the syringe pumps mounted above. In addition, safety constraints were added. The Rl agent could inject a maximum of 1ml per inject with a daily maximum of 4 ml. It also had a mandatory dark period of 10 pm to 6 am.

The third and final iteration separated each compartment into individual devices.

Figure 4: Photo of final hardware setup



System Architecture

The final system is a closed-loop control system. Every thirty minutes the DO probe reads the dissolved oxygen concentration in the culture. The reading is used to construct a five-dimensional vector containing the current DO level, the recent DO trend, the rolling DO average, the remaining daily nutrient budget, and the time of day. The vector was passed through the PPO policy network with two hidden layers of 64 neurons each and ReLU activations to output two actions: (1) LED intensity from 0-100% and nutrient injection from 0 to 1 ml. After 48 steps, the agent ran a gradient update.

Figure 5: System architecture diagram showing decision cycle.

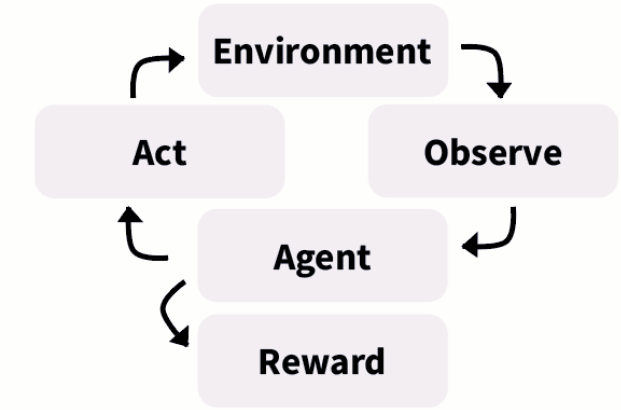
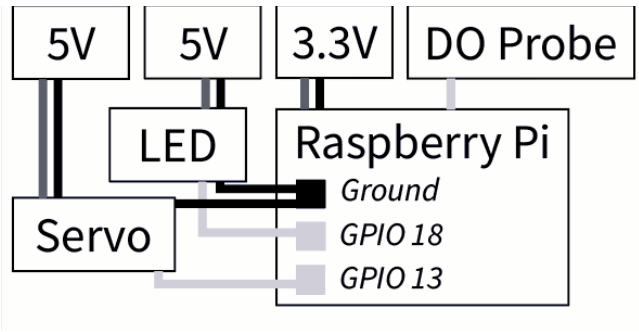


Figure 6: Wiring schematic showing Raspberry Pi connections



Control Conditions

Three control conditions were running across nine reactors, with three per condition. The RL condition used the PPO agent with continual learning described above. The static condition received fixed light at 75% intensity and 2 ml nutrient injections every 12 hours with no sensor feedback. The PID condition used two independent feedback loops, one for light and one for nutrients. They both read the DO every 30 minutes and calculated corrections relative to a 10.5 mg/L setpoint using the standard PID formula:

Equation 8: PID Control Law

$$u\left(t\right) = K_p e\left(t\right) + K_i \int e\left(t\right) dt + K_d \frac{de(t)}{dt}$$

where u(t) is the control output, e(t) is the error between

setpoint and measurement, and Kp, Ki, Kd are the proportional, integral, and derivative gains respectively. Light PID gains were Kp = 2.0, Ki = 0.05, Kd = 0.5. Nutrient PID gains were Kp = 0.08, Ki = 0.002, Kd = 0.01. PID integral terms reset during the dark period to prevent overnight error accumulation. Safety caps were applied to PID as well.

Culture Preparation

Each reactor was prepared identically to ensure the only variable between conditions was the control method. 200 ml of Chlorella vulgaris stock culture was combined with 1 ml of F/2 nutrient concentrate and approximately 49 ml of distilled water to reach a total working volume of 250 ml per reactor. All reactors across each trial were inoculated simultaneously from the same culture batch to ensure identical starting conditions. Initial wet biomass was recorded by weighing each jar.

Experimental Procedure

Three independent five-day trials were conducted per condition. Each trial ran all reactors simultaneously under identical ambient conditions. The RL reactors operated under the continual learning PPO agent described above. The static reactors received fixed light at 75% intensity and 2 ml nutrient injections every 12 hours with no sensor feedback. Dissolved oxygen was logged automatically every 30 minutes via CSV file throughout each trial. Wet biomass was measured daily by removing a 1 ml sample, filtering it, and weighing the filtered material on a digital scale.

At the end of each trial, all reactors were emptied and sterilized with 70% ethanol, rinsed with distilled water, and allowed to dry completely before the next trial. A new culture batch was prepared for each trial to ensure independent starting conditions.

RESULTS

Simulation Training

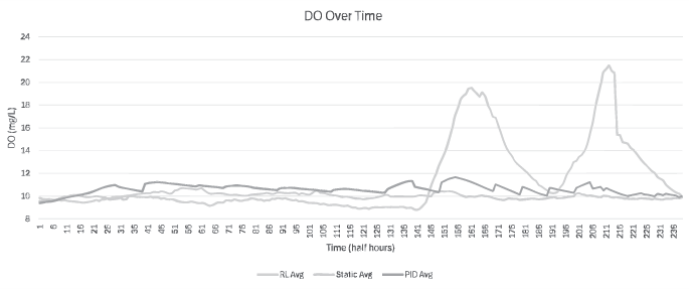
The PPO agent was trained for 200,000 steps in the custom Gymnasium simulation environment. The mean episode reward increased from 1.86 at the start of training to 29.28 at completion. The reward curve stabilized at approximately 130,000 steps, after which no significant improvement was observed, confirming that the policy had converged within the allocated training budget.

Dissolved Oxygen Over Time

Dissolved oxygen was measured every 30 minutes across all nine reactors for each five-day trial, with 239 time steps per trial. During the first three days of each trial, dissolved oxygen levels across RL and static conditions remained similar. Beginning on days four and five, the RL reactors exhibited sharp increases in dissolved oxygen, with trials reaching peak values as high as 23.84 mg/L. The static reactors showed no comparable spike, with a maximum recorded value of 13.92 mg/L across all trials. The PID reactors showed smaller increases consistent with their feedback mechanism, reaching a maximum of 11.90 mg/L. This delayed divergence in RL performance is consistent with the continual learning mechanism: the agent needed multiple real cultivation cycles to update its policy sufficiently before the behavioral changes became visible in the dissolved oxygen data.

Across all three trials, the mean dissolved oxygen for the RL condition was 11.05 mg/L, 10.03 mg/L for the static condition, and 10.60 mg/L for PID, representing improvements of 10.2% and 4.3%.

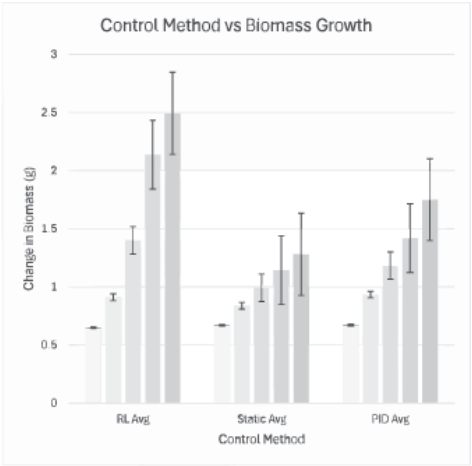
Figure 7: Graph of Dissolved Oxygen over time



Biomass Growth

Wet biomass was measured daily across all nine reactors for each trial. By day five, mean final wet biomass increase was 2.49 g for the RL condition, 1.28 g for the static condition, and 1.75g for PID, representing a 94.7% and 42.8% greater biomass increase.

Figure 8: Biomass over Time Graph



Data Analysis

A two-sample t-test was conducted on per-trial mean dissolved oxygen values and final biomass measurements to assess statistical significance between conditions, using an unequal variance configuration (Type 3).

Figure 9: Mean DO and Biomass Data Table

Metric	RL v Static p	RL v PID p
Mean DO (mg/L)	0.045	0.055
Biomass (g)	0.006	0.002

The standard errors were RL DO ±0.134 mg/L, Static DO ±0.271 mg/L, PID DO ±0.086 mg/L; RL Biomass ±0.049 g, Static Biomass ±0.134 g, PID Biomass ±0.017 g. The non-overlapping standard error bars on biomass confirm visual separation between all three conditions.

The DO result versus static ($p = 0.045$) is statistically significant but borderline with $n = 3$. The DO result versus PID ($p = 0.055$) does not reach the 0.05 threshold. Because dissolved oxygen is susceptible to environmental noise from uncontrolled temperature, biomass was used as the primary performance metric, as it integrates cumulative growth across the full trial rather than reflecting instantaneous conditions. The biomass results at $p = 0.006$ and $p = 0.002$ are the stronger confirmation of the performance difference.

DISCUSSION

The results demonstrate that the continual RL agent outperformed both static and PID control across three independent trials. Mean dissolved oxygen was 10.2% higher than static and 4.3% higher than PID. Final biomass was 94.7% greater than static and 42.8% greater than PID, with statistically significant results on biomass for both comparisons. The most notable pattern in the data was the delayed divergence. All conditions performed similarly for the first three days before the RL reactors spiked sharply on days four and five. This is consistent with the continual learning mechanism needing real cultivation cycles to update the policy.

Two mechanisms most likely explain why RL outperformed static. First, the agent averaged 63% light intensity during daylight hours compared to the static fixed 75%. However, it produced significantly more oxygen. Chlorella vulgaris experiences photoinhibition above 100-150 µmol photons $m^{-2}s^{-1}$, meaning the static reactor's constant 75% likely triggered suppression of photosynthesis while the agent's Haldane-trained policy naturally avoided this zone. Second, the agent's nutrient behavior changed over the course of the trial. It started by front-loading injections early in the day and by days three and four had shifted to distributing them across peak photosynthetic hours.

The DO t-test versus PID ($p = 0.055$) did not reach significance, which is expected given that PID also uses sensor feedback and thus closes some of the gap that exists between RL and purely static control. The significant biomass result versus PID ($p = 0.002$) suggests the RL advantage is primarily driven by the coupled light-nutrient optimization. The high static trial 2 mean (10.58 mg/L versus 9.77 and 9.76 in the other two trials) is most likely explained by uncontrolled temperature variation and highlights the need for temperature monitoring in future work. Other limitations include using wet mass as a biomass proxy rather than dry weight filtration, and the 250 ml lab scale not replicating the light attenuation and mixing dynamics of industrial photobioreactors.

However, the results still establish clear proof that continual reinforcement learning on embedded hardware can achieve statistically significant improvements in microalgae cultivation over both static and PID control. At industrial scale, this translates to meaningfully more biomass, biofuel, and carbon captured per reactor. Future work should add temperature as a third controlled variable, conduct a formal RL versus tuned PID comparison at multiple setpoints, and scale to a 5–20-liter pilot reactor for industrial validation

REFERENCES

Ameer Ali Kubar, Mehmood, S., Schagerl, M., Kumar, S., Hu, X., Zhu, F., Xu, X., Ni, J., & Huo, S. (2025). Integrated hybrid Nested-bottled photobioreactor for enhanced mixing, mass transfer, and CO2 fixation in Arthrospira platensis raceway pond cultivation systems. Biotechnology for Biofuels and Bioproducts, 18(1). https://doi.org/10.1186/s13068-025-02670-1

Arif Gasilov. (2025, May 7). Groundbreaking Study Confirms Algae Can Capture Carbon 50x Faster Than Trees, Offering a Powerful Climate Solu. Seaside. https://www.seasidesustainability.org/post/groundbreaking-study-confirms-algae-can-capture-carbon-50x-faster-than-trees-offering-a-powerful-cl

Chisti, Y. (2007). Biodiesel from microalgae. Biotechnology Advances, 25(3), 294–306. https://doi.org/10.1016/j.biotechadv.2007.02.001

Deprá, M. C., Mérida, L. G. R., de Menezes, C. R., Zepka, L. Q., & Jacob-Lopes, E. (2019). A new hybrid photobioreactor design for microalgae culture. Chemical Engineering Research and Design, 144, 1–10. https://doi.org/10.1016/j.cherd.2019.01.023

Garodia, H., Jain, R., & Kini, S. (2025). Optimization of Carbon Capture Rates of Chlorella Vulgaris Using Experimental Testing. Studies in Systems, Decision and Control, 231–242. https://doi.org/10.1007/978-3-031-84628-1_19

Handa, H. (2007). Evolutionary Computation on Multitask Reinforcement Learning Problems. 685–688. https://doi.org/10.1109/icnsc.2007.372862

Islam, M. H., & Patlolla, S. R. (2025). A Review on the State-of-the-Art and Commercial Status of Carbon Capture Technologies. Energies, 18(15), 3937. https://doi.org/10.3390/en18153937

Jiao, Z., & Oh, J. (2019). Asynchronous Multitask Reinforcement Learning with Dropout for Continuous Control. 2019 18th IEEE International Conference on Machine Learning and Applications (ICMLA). https://doi.org/10.1109/icmla.2019.00099

Luca, L., Barbu, M., Ifrim, G., Ceanga, E., Miron, M., & Sergiu Caraman. (2018). Fuzzy Control of a Microalgae Grow-Mh Process in Photobioreactors. 26, 480–485. https://doi.org/10.1109/icstcc.2018.8540698

Ma, Z., Cheah, W. Y., Ng, I-Son., Chang, J.-S., Zhao, M., & Show, P. L. (2022). Microalgae-based biotechnological sequestration of carbon dioxide for net zero emissions. Trends in Biotechnology, 40(12), 1439–1453. https://doi.org/10.1016/j.tibtech.2022.09.002

Mania, H., Guy, A., & Recht, B. (2018). Simple random search provides a competitive approach to reinforcement learning. ArXiv:1803.07055 [Cs, Math, Stat]. https://arxiv.org/abs/1803.07055

Mulhern, O. (2020, August 12). A 4.5 Billion-Year History of CO2 in our Atmosphere. Earth.org - Past | Present | Future; Earth.org. https://earth.org/data_visualization/a-brief-history-of-co2/

Ugwu, C. U., & Aoyagi, H. (2011). Microalgal Culture Systems: An Insight into their Designs, Operation and Applications. Trends in Applied Sciences Research, 6(10), 1222–1229. https://doi.org/10.17311/tasr.2011.1222.1229

Zocco, F., García, J., & Haddad, W. M. (2025). Circular Microalgae-Based Carbon Control for Net Zero. ArXiv.org. https://arxiv.org/abs/2502.02382